

Home Assignment-week1day1

1) Classification Algorithm

a) Age <30 and Govt satisfaction less ,urban

Age-40 , Govt. satisfaction-0.600/10 , Urban voter-Yes.

Party A=0.0%

Party T=97.4%

Party C=2.4%

Undecided=0.2%

Party T

11.9 pts $\leftarrow$ base points

$- 0.149 \times 40.0$ (Age) = -5.96

$- 1.49 \times 0.600$ (Govt. satisfaction) = -0.893

$+ 2.15 \times \text{Yes}$ (Urban voter) = +2.15

= 7.19 pts

b) Age >50 and Govt satisfaction less ,non urban

Age-59 , Govt. satisfaction-0.800/10 , Urban voter-No.

Party A=0.0%

Party T=4.0%

Party C=95.7%

Undecided=0.3%

Party C

1.20 pts $\leftarrow$ base points

$+ 0.078 \times 59.0$ (Age) = +4.61

$- 0.900 \times 0.800$ (Govt. satisfaction) = -0.720

$$- 0.285 \times \text{No (Urban voter)} = +0$$

$$= 5.09 \text{ pts}$$

=====

2) Linear regression

Check the Calculation for price and time

a) QA Salary

Base salary — the starting value, before any feature counts

3.10 LPA

Experience

$$0.966 \text{ LPA per year} \times 5.00 \text{ yrs}$$

$$= +4.83 \text{ LPA}$$

Skill rating

$$1.34 \text{ LPA per point} \times 6.50 / 10$$

$$= +8.72 \text{ LPA}$$

TOTAL 16.7 LPA

b) Food Delivery time

Base time — the starting value, before any feature counts

7.56 min

Distance

$$2.40 \text{ min per km} \times 3.00 \text{ km}$$

$$= +7.21 \text{ min}$$

Traffic

3.16 min per point × 4.50 /10

= +14.2 min

TOTAL 29.0 min